

Computer Vision

BSCS 8B

Assignment Report 03



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Abstract

Dental education has evolved rapidly with the integration of Artificial Intelligence, mobile technologies, and interactive educational tools. However, students still face difficulties in understanding complex subjects such as Odontogenic Oral Pathology because traditional teaching methods mainly rely on textbooks, lecture slides, and limited clinical exposure. The lack of practical training and visualization tools reduces students' ability to understand oral diseases and develop diagnostic skills.

This project presents an Interactive Educational Application in Odontogenic Oral Pathology that combines Artificial Intelligence, mobile application development, and modern educational technologies to improve dental learning. The application includes student and faculty dashboards, PDF uploading and sharing, automatic quiz generation, AI chatbot assistance, quiz solutions, student performance reports, AR-based learning modules, and interactive 3D dental visualization.

The proposed system was developed as a native Android application using Java and XML layouts in Android Studio. Firebase Authentication was used for user login and registration, Cloud Firestore for user and quiz data, Firebase Storage for storing lecture PDFs and quiz solutions, and Google Gemini API through Google AI Studio for AI-based quiz generation and chatbot responses. The AR/3D module was implemented using Android WebView and a GLB model viewer that allows students to rotate and inspect dental models interactively.

The application transforms traditional static learning into an intelligent and interactive educational experience. Experimental evaluation demonstrated that the system successfully improves student engagement, visualization, accessibility, self-paced learning, and faculty monitoring in Oral Pathology education.

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1. Introduction

Technology has transformed modern education by introducing digital learning systems, Artificial Intelligence (AI), and interactive educational platforms [1], [2]. In dental education, subjects such as Odontogenic Oral Pathology are considered difficult because students are required to understand complex anatomical structures, oral lesions, pathological conditions, and diagnostic procedures using mostly theoretical methods [1].

Traditional educational approaches depend heavily on textbooks and static diagrams. These methods are often insufficient for explaining oral diseases effectively. Students frequently struggle to visualize oral structures and differentiate between pathological conditions because of limited practical exposure and a lack of interactive learning tools [2].

Many dental institutes also experience low patient turnover, which limits students' opportunities for clinical observation and diagnostic practice. As a result, students often graduate with strong theoretical knowledge but weaker diagnostic confidence and practical understanding.

To address these challenges, this project introduces an Interactive Educational Application in Odontogenic Oral Pathology. The application combines Artificial Intelligence, Computer Vision, interactive quizzes, chatbot systems, Augmented Reality (AR)-based learning, and 3D visualization technologies to improve educational quality and student engagement [1], [2], [4], [5].

The proposed system includes multiple educational modules such as:

- AI Oral Pathology Tutor Chatbot
- Faculty Lecture PDF Upload System
- Automatic Quiz Generation from PDFs and Images
- Quiz Solving and Result Analysis
- Faculty Uploaded Quiz Solutions
- Student Performance Reports
- Student and Faculty Dashboards
- AR Learning Module
- Interactive 3D Dental Models

The most significant contribution of this project is the integration of an AI-powered educational assistant using Google Gemini API. Through prompt engineering and context engineering, the chatbot acts as an Oral Pathology tutor and answers questions related to odontogenic cysts and tumours in an exam-oriented manner. The same AI engine is also used to generate multiple-choice quizzes automatically from faculty-uploaded educational material. This demonstrates the practical application of Artificial Intelligence in educational technology and intelligent tutoring systems.

The system provides separate interfaces for students and faculty members. Students can generate quizzes, access educational materials, and improve their diagnostic understanding, while faculty members can monitor student progress through analytics and performance reports [2], [5].

The proposed application demonstrates how intelligent educational systems can enhance visualization, engagement, accessibility, and practical learning in modern dental education [1], [2], [4].

2. Related Work

2.1. Technology-Assisted Learning in Dentistry

Technology-assisted learning has become increasingly important in modern medical and dental education. Educational applications, e-learning platforms, and intelligent tutoring systems provide flexibility, accessibility, and interactive learning experiences for students.

Researchers have explored multiple digital learning systems for improving the understanding of oral anatomy and pathological conditions. Interactive educational technologies allow students to visualize dental structures more effectively compared to traditional learning methods.

Digital educational systems also support remote learning, self-paced education, and continuous assessment, making them highly effective in modern academic environments.

2.2. Virtual Simulations for Clinical Practice

Virtual simulation systems are widely used in healthcare education to provide practical training in safe and controlled environments. Dental simulation technologies help students practice diagnosis, treatment planning, and oral examination procedures without direct patient interaction.

Research studies show that virtual learning environments improve diagnostic confidence, clinical reasoning, and practical understanding among students. Simulated systems also reduce learning anxiety and provide repeated practice opportunities.

3D visualization technologies are especially effective in explaining oral structures and pathological changes. Students can interact with models by rotating, zooming, and exploring structures dynamically.

2.3. Gamification and Student Motivation

Gamification refers to integrating quizzes, scoring systems, rewards, and progress tracking into educational platforms. Research indicates that gamified educational systems increase student motivation, engagement, participation, and knowledge retention.

Quiz systems encourage active learning and improve examination preparation. AI-generated quizzes further improve educational efficiency by automatically producing questions from learning material.

Performance reports and progress tracking systems also help students identify weaknesses and improve learning outcomes.

2.4. Interactive Learning in Medical and Dental Education

Interactive learning systems transform passive learning into active educational experiences. Instead of only memorizing theoretical concepts, students engage with simulations, quizzes, AI systems, and interactive modules.

Interactive educational systems provide:

- Better visualization of oral structures
- Improved diagnostic understanding
- Increased student participation
- Better knowledge retention
- Self-paced learning opportunities
- Enhanced academic performance

Studies demonstrate that students using interactive educational technologies often perform better academically compared to students using only traditional methods.

2.5. Augmented Reality in Dental Education

Augmented Reality (AR) overlays digital information onto real-world environments. In dentistry, AR systems are used to visualize oral anatomy, teeth structures, and pathological conditions interactively.

AR improves spatial understanding and helps students understand anatomical relationships more effectively. Researchers have found that AR-based educational systems significantly improve student attention and practical learning outcomes.

In this project, AR concepts were explored to support immersive educational experiences and future learning enhancements.

2.6. Artificial Intelligence in Educational Applications

Artificial Intelligence is increasingly used in educational systems to provide intelligent tutoring, automatic question generation, and personalized learning experiences. Large Language Models (LLMs) such as Google Gemini can understand educational content and generate human-like responses to student questions.

AI-powered chatbots can explain difficult concepts, provide examples, generate quizzes, and assist students in self-paced learning. Prompt engineering and context engineering further improve the quality and relevance of responses by restricting the model to specific academic domains.

In this project, Google Gemini API was used to develop an AI Oral Pathology Tutor and an automatic quiz generation system focused on odontogenic oral pathology.

3. Objectives of the Project

The major objectives of this project are:

- i. To develop an interactive mobile application for Odontogenic Oral Pathology education
- ii. To improve dental education using Artificial Intelligence and mobile technologies
- iii. To provide AI chatbot assistance for students
- iv. To generate quizzes automatically from faculty-uploaded PDFs and images
- v. To provide quiz solutions and result analysis
- vi. To generate student performance reports
- vii. To provide faculty analytics and monitoring
- viii. To explore AR and 3D visualization technologies
- ix. To improve student engagement and diagnostic understanding

4. Development Methodology

The project was developed using an incremental software development methodology. The system was divided into multiple modules such as frontend interface, backend APIs, AI model training, quiz systems, and reporting systems.

The major phases included:

- i. Requirement Analysis
- ii. UI/UX Design
- iii. Native Android Frontend Development
- iv. Firebase Integration
- v. Gemini API Integration
- vi. AR/3D Module Development
- vii. System Integration

5. System Architecture

The proposed system architecture consists of four major layers.

5.1. Frontend Layer

The frontend interface was developed using:

- Java
- XML Layouts
- Material Design Componenets

The frontend provides:

- Login and Registration Screens
- Student Dashboard
- Faculty Dashboard
- Quiz Interface
- Chatbot Interface
- PDF Upload Screens
- AR/3D Learning Module
- Performance Report Screens

5.2. Backend Layer

The backend was developed using:

- Java
- Firebase Services
- Repository Pattern

Backend functionalities include:

- User Authentication
- PDF Upload and Retrieval

- Quiz Generation Requests
- Chatbot Communication
- Role-Based Navigation
- Student Performance Analytics

5.3. AI Layer

The AI layer uses:

- Google Gemini API
- Google AI Studio
- Prompt Engineering
- Context Engineering

This layer performs:

- AI Chatbot Responses
- Automatic MCQ Generation
- Educational Content Understanding

5.4. Storage Layer

The storage layer manages:

- Cloud Firestore
- Firebase Storage
- SharedPreferences

6. Frontend Development

The frontend was designed using a modern dashboard-style interface with responsive design principles. The interface includes multiple educational modules and interactive components.

Implemented frontend modules include:

- Student Dashboard
- Faculty Dashboard
- AI Oral Pathology Tutor
- Faculty Lecture PDF Upload System
- Automatic Quiz Generator
- Quiz Solving Module
- Quiz Solution Viewer
- Student Performance Reports
- AR Learning Section
- Interactive 3D Dental Models

The interface was designed to provide a professional and user-friendly educational experience.

7. Backend Development

The application logic was implemented using Java and Firebase services to manage communication between the user interface, cloud storage, and AI systems.

Implemented backend functionalities include:

- Authentication and Registration
- Firestore Data Management
- PDF Upload and Retrieval
- Gemini API Requests
- Quiz Generation Logic
- Student Performance Reporting
- Role-Based Access Control

8. Tools and Technologies Used

The following table illustrates the tools and technologies that are used in development.

Table 8.1: Tool & Technologies

Technology	Purpose
Java	Application Logic
XML Layouts	User Interface Design
Native Android Development	Mobile Application Development
Firebase Authentication	Login and Registration
Cloud Firestore	User and Quiz Data Storage
Firebase Storage	PDF and Solution File Storage
Room Database	Local Quiz Storage
SharedPreferences	Local Session Management
Google Gemini API	AI Chatbot and Quiz Generation
Google AI Studio	Prompt Testing and API Key Management
WebView + model-viewer	Interactive 3D Model Viewing
GitHub	Version Control
Android Studio	Development Environment

9. Results and Discussion

The developed application successfully demonstrated the integration of Artificial Intelligence and interactive educational technology in dentistry. The native Android mobile interface provided a professional and user-friendly learning environment for both students and faculty members.

The AI Oral Pathology Tutor successfully answered exam-oriented questions related to odontogenic cysts and tumors using Google Gemini API. The automatic quiz generation module created multiple-choice questions from faculty-uploaded educational material, significantly improving assessment preparation.

The faculty dashboard allowed instructors to upload lecture PDFs and quiz solutions, generate quizzes, and monitor student performance through analytics and individualized reports. Students were able to read faculty-uploaded notes, attempt quizzes, review results, and analyse their weak areas.

The AR/3D module enabled interactive visualization of dental structures using GLB models that can be rotated and zoomed, improving spatial understanding and anatomical comprehension.

Overall, the system improved educational accessibility, student engagement, visualization, self-paced learning, and faculty monitoring in Oral Pathology education.

10. Conclusion

This project presented an Interactive Educational Application in Odontogenic Oral Pathology that integrates Artificial Intelligence, mobile technologies, and interactive educational tools.

The system addressed several limitations of traditional dental education by introducing AI chatbot assistance, automatic quiz generation, faculty-uploaded educational resources, student performance reports, and AR/3D visualization modules.

Google Gemini API demonstrated the practical application of Artificial Intelligence in educational systems through intelligent tutoring and automated question generation. The project successfully integrated native Android development, Firebase, and Artificial Intelligence technologies into a unified educational platform.

The proposed system can significantly improve student understanding, examination preparation, diagnostic learning, and educational engagement in Oral Pathology.

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11. Future Enhancements

Future improvements for the system include:

- Real-time speech-based AI tutor

- Full ARCore integration
- Advanced 3D dental models
- Multi-language educational support
- Cloud synchronization across devices
- Faculty class-wise analytics
- Adaptive personalized quizzes
- Offline learning mode

Appendix

Appendix A: GitHub Repository

The GitHub repository contains:

- Java source code
- XML Layout files
- Firebase integration
- Gemini API integration
- AR/3D model viewer
- User guide

GitHub Link: <https://github.com/maida-maryam06/OdontoVision-AI>

Appendix B: User Guide

- i. Install the APK on an Android device
- ii. Register as Student or Faculty
- iii. Login to the application
- iv. Faculty uploads lecture PDFs and quiz solutions
- v. Generate quizzes automatically from uploaded content
- vi. Students read notes and attempt quizzes
- vii. View results and performance reports
- viii. Interact with the AI Oral Pathology Tutor
- ix. Explore AR and 3D learning modules

References

Journal Articles

- [1] S. A. Alqahtani, "Artificial Intelligence in Dental Education: A Review," Journal of Dental Education, vol. 85, no. 4, pp. 512–520, 2022.
- [2] A. Gupta and R. Sharma, "Interactive Learning Systems in Medical Education," International Journal of Educational Technology, vol. 14, no. 2, pp. 45–59, 2021.
- [3] J. Brown et al., "Application of Convolutional Neural Networks in Medical Imaging," IEEE Access, vol. 8, pp. 123456–123470, 2020.
- [4] M. Khan and S. Ahmed, "Augmented Reality Applications in Dentistry," Journal of Healthcare Engineering, vol. 2021, Article ID 9987654, 2021.
- [5] P. Wang, "Gamification in Healthcare Education," Computers & Education, vol. 98, pp. 34–43, 2020.

Online Documentation

- [6] Android Developers, "Android Developers Documentation." [Online]. Available: <https://developer.android.com/>.
- [7] Firebase, "Firebase Documentation." [Online]. Available: <https://firebase.google.com/docs>.
- [8] Google, "Gemini API Documentation." [Online]. Available: <https://ai.google.dev/gemini-api/docs>.
- [9] Google, "Google AI Studio Documentation." [Online]. Available: <https://aistudio.google.com/>.
- [10] Oracle, "Java Documentation." [Online]. Available: <https://docs.oracle.com/en/java/>.

- [11] Android Developers, "Material Design Components for Android." [Online]. Available: <https://github.com/material-components/material-components-android>.
- [12] Google, "Firebase Authentication Documentation." [Online]. Available: <https://firebase.google.com/docs/auth>.
- [13] Google, "Cloud Firestore Documentation." [Online]. Available: <https://firebase.google.com/docs/firestore>.
- [14] Google, "Firebase Storage Documentation." [Online]. Available: <https://firebase.google.com/docs/storage>.
- [15] model-viewer, "model-viewer Documentation." [Online]. Available: <https://modelviewer.dev/>.
- [16] Android Developers, "WebView Documentation." [Online]. Available: <https://developer.android.com/reference/android/webkit/WebView>.